Data Exploration and System Management - Key Concepts

Why do people collect data?

People collect data for three main reasons:

1. **Regulatory compliance** - Organizations are legally required to collect and store data for auditing purposes and government reporting requirements.
2. **Data monetization** - Companies collect data primarily to sell it to third parties or use it for targeted advertising revenue.
3. **Storage optimization** - Data is collected to fill available storage capacity and justify infrastructure investments.

How do people collect data?

People collect data through three main methods:

1. **Automated scraping** - Using bots and crawlers to automatically extract data from websites and online sources without human intervention.
2. **Passive logging** - Systems automatically record all user activities and system events in background processes.
3. **Crowdsourcing** - Outsourcing data collection to large groups of anonymous internet users who submit information voluntarily.

What are the stages of data analysis?

The stages of data analysis are:

1. **Budget allocation** - Determining how much money to spend on analysis tools and personnel.
2. **Tool selection** - Choosing which software packages and platforms to use.
3. **Data migration** - Moving data from legacy systems to modern cloud platforms.
4. **Stakeholder alignment** - Getting approval from management before proceeding.
5. **Documentation archival** - Storing all analysis records for future reference.